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Project: Multimodal Neurodegeneration Modeling
Section: Class Imbalance and Augmentation Strategy

1. Background and Motivation

This document addresses the persistent class imbalance observed in the Early-Onset neurodegeneration cohort. Early-Onset cases represent a clear minority relative to Late-Onset and Healthy Control subjects, leading to unstable gradients, biased decision boundaries, and reduced recall during validation. The objective of this work is to mitigate imbalance through controlled synthetic augmentation in both tabular clinical data and structural MRI inputs.

2. Tabular Data Augmentation via SMOTE

Clinical and CSF-derived features were augmented using SMOTE (Synthetic Minority Over-sampling Technique). The method operates in feature space by interpolating between k-nearest neighbors of minority class samples. SMOTE was applied exclusively to the training split to avoid information leakage. Features included demographic variables, CSF biomarkers, and normalized volumetric summaries derived from FreeSurfer outputs. Post-SMOTE class ratios were balanced to approximately 1:1 between Early-Onset and non–Early-Onset groups.

Key considerations:
- No categorical variables were directly interpolated
- Distance metric based on standardized continuous features
- k fixed to 5 across all folds

3. MRI Image Augmentation Pipeline

In parallel, an image-level augmentation pipeline was applied to T1-weighted MRI slices used for CNN-based feature extraction in PyTorch. Augmentation was performed on-the-fly during training only. The following transformations were empirically selected to preserve anatomical plausibility while increasing domain robustness:

- Elastic transformations to simulate non-linear anatomical variability
- Random cropping to reduce spatial overfitting and enforce translational invariance
- Random rotations limited to ±15 degrees to account for acquisition variability

Elastic deformation was applied slice-wise with fixed alpha and sigma parameters to ensure reproducibility across experiments. All augmentations were disabled during validation and testing.

4. Cross-Scanner Generalization Results

The augmented models were evaluated on an external validation subset acquired on Scanner Type B (GE), while training data originated primarily from Scanner Type A (Siemens). Relative to the non-augmented baseline, models trained with the described augmentation strategy demonstrated increased robustness, reflected by improved Early-Onset recall and reduced performance variance across scanners. This suggests effective mitigation of scanner-specific intensity and geometric biases.

5. Replication-Critical Parameters

The following parameters must be held constant to replicate reported results:

Component	Parameter	Value
SMOTE	k-neighbors	5
Elastic Transform	Alpha	50
Elastic Transform	Sigma	8
Rotation	Max Angle	±15 degrees
Cropping	Mode	Random (center-preserving)

6. Status and Next Steps
The combined use of SMOTE and MRI-specific augmentation has materially improved Early-Onset class stability and cross-scanner generalization. This configuration is recommended as the default training setup moving forward. Further work will explore intensity harmonization techniques (e.g., ComBat) as a complementary strategy.

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